Software Quality and Testing

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Overview of Software Testing

- Software testing evaluates whether software behaves according to requirements.
- It ensures reliability, usability and security before release.
- Testing helps detect defects early in development.
- Quality software improves user trust and system stability.

Why Testing is Important

- Software errors can cause financial loss and safety risks.
- Testing reduces bugs before software reaches customers.
- It improves performance and reliability of applications.
- Organizations rely on testing to maintain product quality.

Real World Example

- A banking application incorrectly calculates transaction balances which could result in financial losses for customers.
- The issue may occur due to logical errors or incorrect data handling.
- Proper testing would detect this before system deployment.

Understanding Software Quality

- Software quality measures how well a system meets user expectations.
- High quality systems are reliable, efficient and secure.
- Quality also includes maintainability and usability.
- Testing plays a key role in maintaining these attributes.

Key Terminology

- **Error** – mistake made by developers during coding or design.
- **Fault** – defect present inside the software system.
- **Failure** – incorrect output produced by the software.
- **Incident** – observable event indicating a system problem.

Types of Software Testing

- ***Unit Testing*** – It includes testing individual components.
- ***Integration Testing*** – It verify interaction between modules.
- ***System Testing*** – In this we test the entire application.
- ***Regression Testing*** – ensuring updates do not introduce new defects.

Verification vs Validation

***Verification** – Its like - are we building the product correctly?*

Verification checks that **software is developed accordingly to the design specifications and requirements**.

- Focus **process and development stage**
- Ensures the system is built **correctly**
- Performed through **reviews, inspections and walkthroughs**
- Done **before execution of the software**

Example :

Checking whether that the login modules is implemented according to the system design document.

***Validation** – Its like - are we building the right product?*

Validation checks whether the **final software product meets the users need and expectation**.

- Focuses on **actual functionality of the system**
- Ensures the system solves the **real user problem**
- Performed through **testing and user acceptance testing**
- Done **after the software is developed**
- **Example:**

  Testing whether users can successfully log into the system using correct credentials.

Use Case Example – Login System

Use case shows how user interact with system to complete a specific task. It helps testers understand system behaviour and design appropriate test cases.

Use Case: User Login

Actor : Registered User

Goal : Access their account in the system.

Main Steps

1. The user opens the login page.
2. Then enters a valid username and password.
3. Then the system verifies the credential.
4. If the credentials are correct only then the user ican access to their account dashboard.

Alternative Scenarios

- If the **password is incorrect then** the system shows the error message.
- If the **fields are empty**, the systm asks the user to enter the required information.
- After **multiple fail attempts** the account may be temporarily locked for security reason.

Testing Perspective

Testers create different **test cases** to verify:

- Successful login with valid credentials
- Error handling for invalid inputs
- System behaviour during repeated failed login attempts

Testing Tools

1. Selenium

- It is atomation testing tool for web applications.
- It Supports various programming languages like Java, Python and C#.
- Allows testers to run tests across different browsers and operating systems.

2. JUnit

- It is **testing framework for Java** applications.
- It also helps developers write and run automated test cases for program components.
- Commonly used during the **development stage**.

3. TestNG

- It is advanced testing framework by JUnit.
- Supports **parallel testing, test grouping, and detailed reports**.
- Frequently used for **large-scale Java testing projects**.

4. Ranorex

- It is **automated testing tool for deskop, web and mobile applications**.
- Supports **recording and playback of test cases**.

Why Testing Tools Are Important

- Reduce manual testing effort
- Improve testing speed and accuracy
- Support automated testing processes
- Help identify defects quickly

Reflection on Learning

- This week learning introduced the importance of **software quality and testing** in the software development process. I learned that testing is not only performed after coding but should be integrated throughout the **entire software development life cycle**.

- Understanding concepts such as **errors, faults, failures, and testing levels** helped me realise how software defects can impact real-world systems. Real-life examples of software failures highlighted the importance of proper testing in preventing financial losses, safety risks, and system breakdowns.

- I also understood **different testing methods like unit testing, integration testing and system testing** in ensuring reliable software. These concepts will be useful in real-world software development, as developers must test applications carefully to deliver high-quality and dependable systems.

- Overall, this improved my understanding of how **testing contributes to building secure, reliable and efficient software applications**.

SDLC Process

Requirements

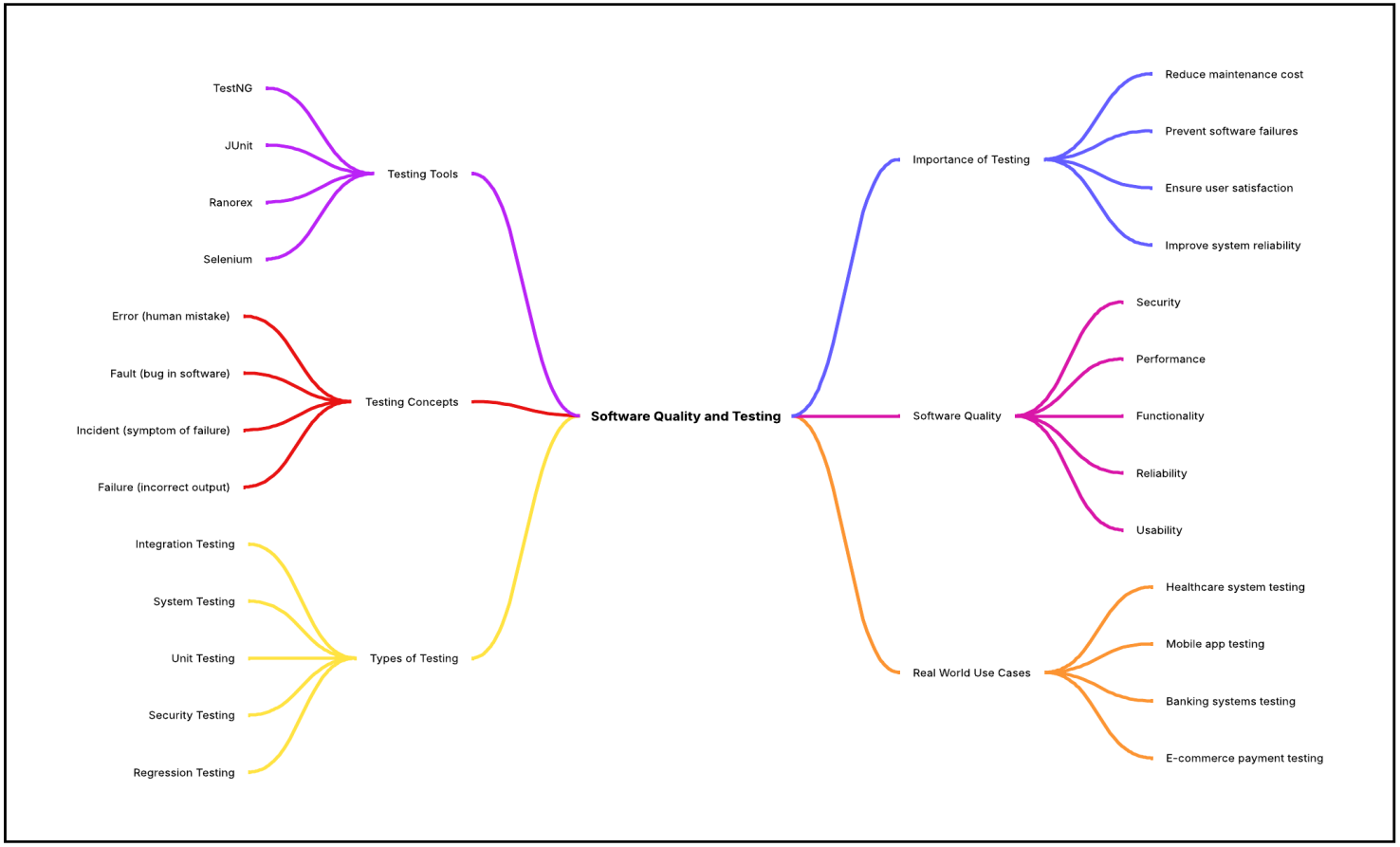
Design

Development

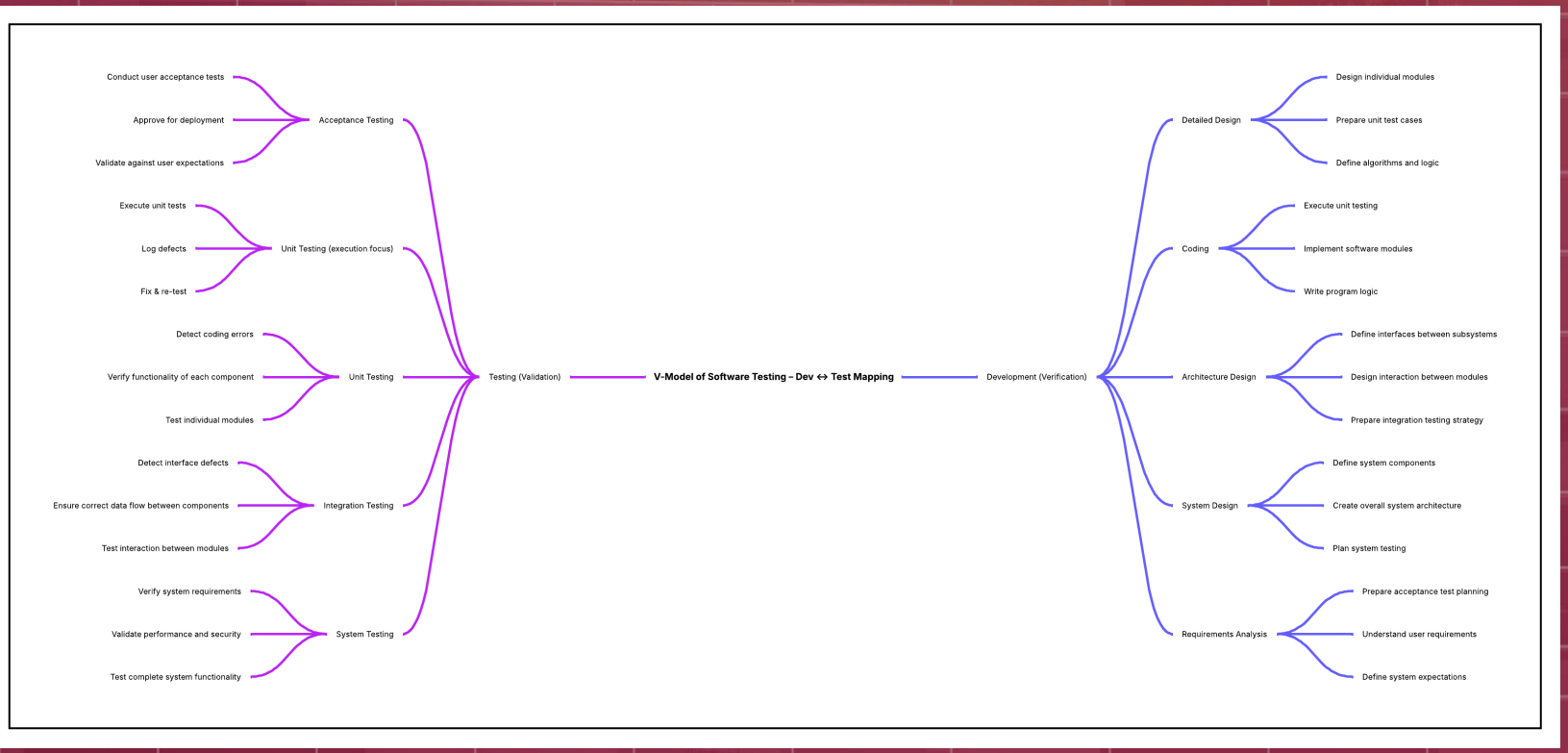
Testing

Deployment

Mind Map : Software Quality & Testing



V Model of Testing



Error → Fault → Failure Flow

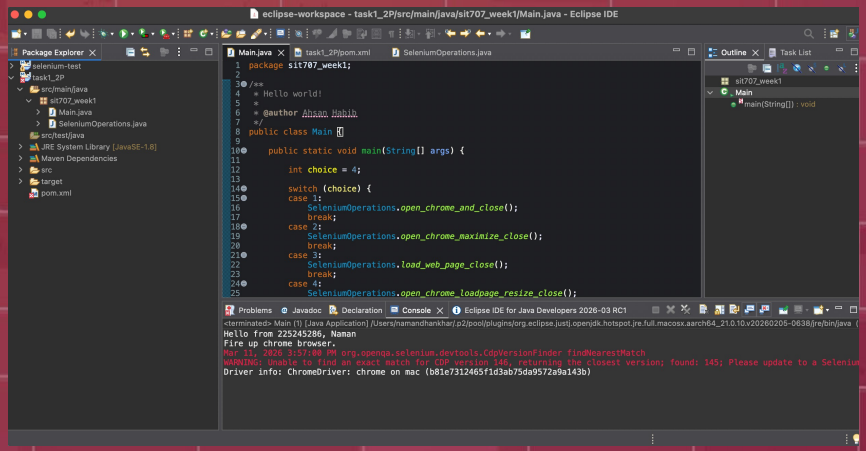
Human Error

Code Fault

System Failure

User Incident

Real Evidences of Learning



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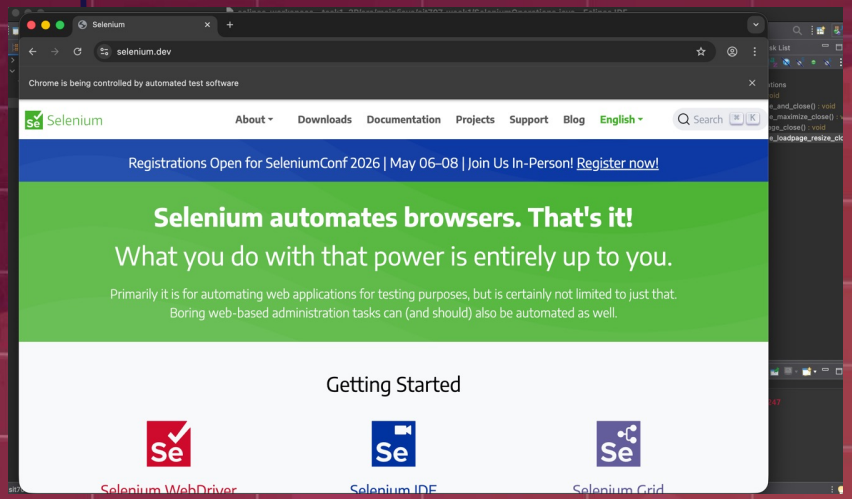

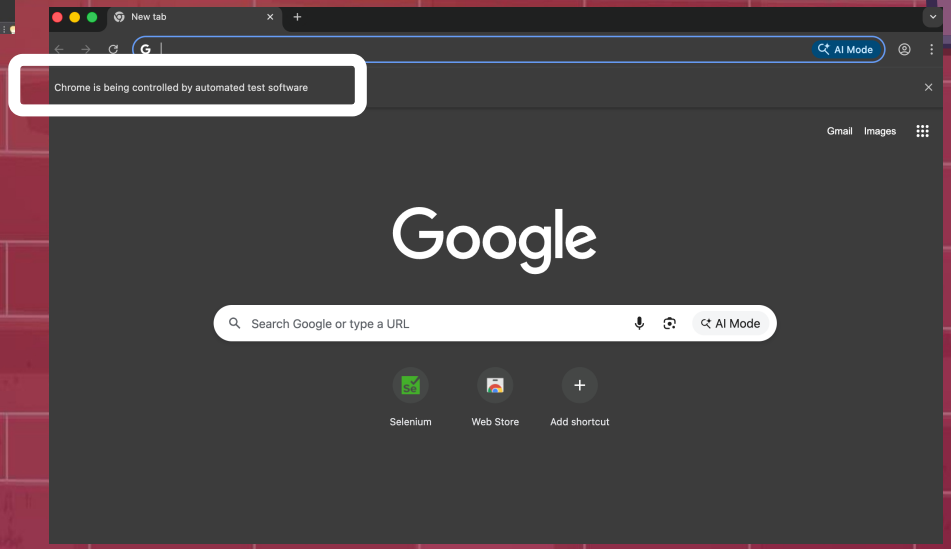

Chrome for Testing availability

This page lists the latest available cross-platform Chrome for Testing versions and assets per Chrome release channel.

Consult our JSON API endpoints if you're looking to build automated scripts based on Chrome for Testing release data.

Last updated @ 2026-03-15T11:12:37.115Z

Channel	Version	Revision	Status
Stable	146.0.7680.80	r1582197	✅
Beta	147.0.7727.3	r1596535	✅
Dev	148.0.7730.2	r1598053	✅
Canary	148.0.7733.0	r1599001	✅

Stable

Version: 146.0.7680.80 (r1582197)

Binary	Platform	URL	HTTP status
chrome	linux64	https://storage.googleapis.com/chrome-for-testing-public/146.0.7680.80/linux64/chrome-linux64.zip	200
chrome	mac-arm64	https://storage.googleapis.com/chrome-for-testing-public/146.0.7680.80/mac-arm64/chrome-mac-arm64.zip	200
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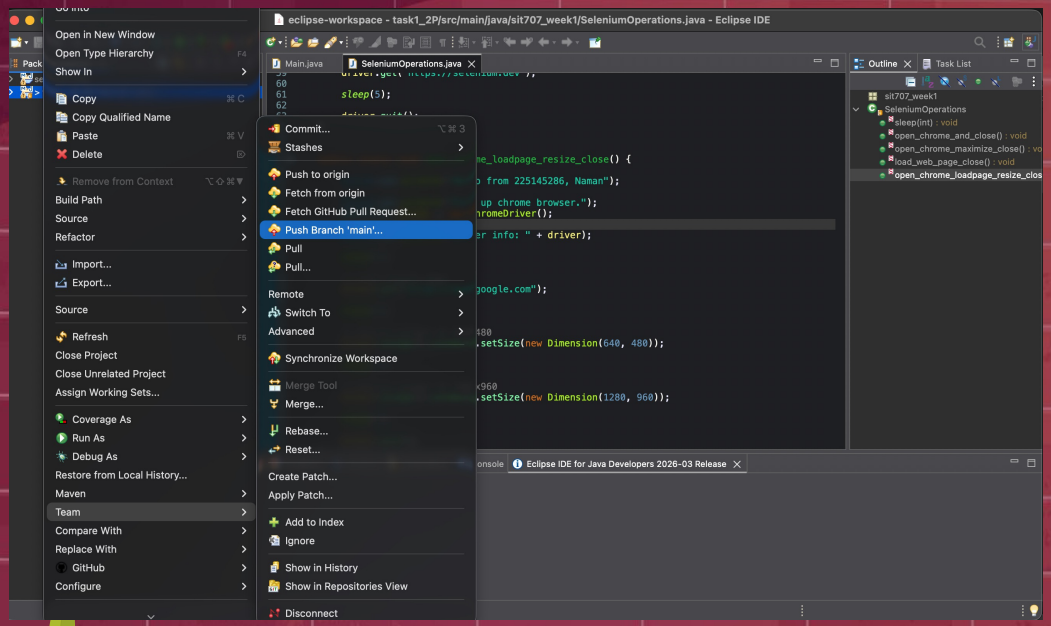
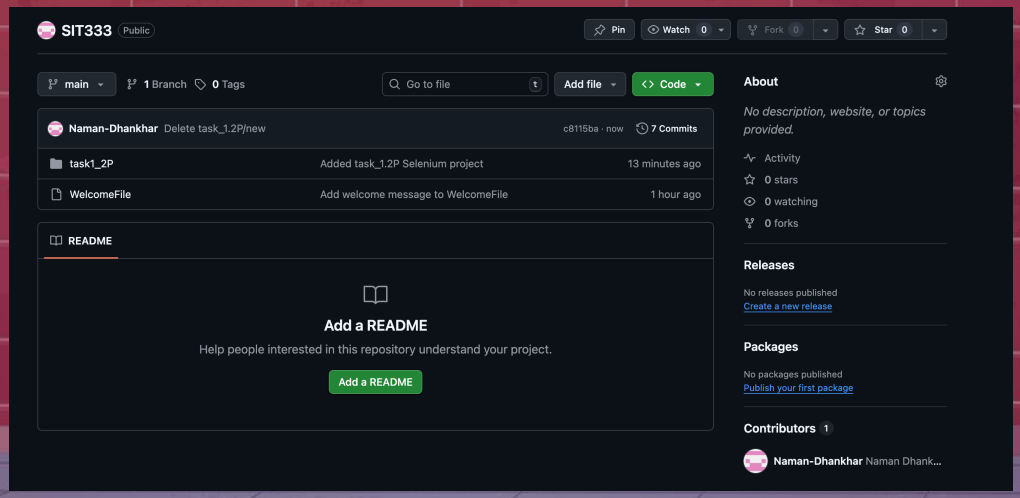
Stable Crome Driver For Testing

Pushing to Git Hub





THANK
YOU !